The Beveridge Curve Analysis

To begin predicting our findings, let's revisit the key conclusions from our last discussion. A significant takeaway was that we are currently in a phase of uncertainty regarding unemployment trends. The critical question is: what trajectory will unemployment take moving forward? Will it continue following the post-pandemic trends we're observing? Will it revert to the pre-pandemic patterns, assuming the economy has recovered fully from the COVID-19 shock? Or could a completely new trend emerge? Before diving into these questions, we need a solid strategy to approach the core issue:

What happens if we reduce the impact of the COVID-19 shock?

The most straightforward method is to remove the values corresponding to the pandemic period. This could potentially clarify the long-term trends by eliminating the noise introduced by the pandemic. However, this approach presents a significant caveat: the pandemic period is not just a set of anomalies but a critical economic event. Ignoring these values outright would strip away important context because the pandemic must be understood as both a shock to the system and a recovery process. The challenge lies in defining the timeline for recovery. How do we determine when the economy fully recovered from the pandemic's impact? The period of recovery is ambiguous, and removing data without a clear rationale could lead to oversimplification. Removing too many points risks losing valuable information, while retaining too many might obscure the underlying trends.

A Balanced Approach: Smoothing the Impact

One way to tackle this is to smooth the weights assigned to the pandemic shock and recovery phases. Rather than entirely removing the data, we could adjust their influence on the analysis based on how far these values deviate from the overall trend. For example, severe outliers during the height of the pandemic (e.g., sudden unemployment spikes) could be given lower weights. Recovery data could be gradually integrated into the trend as the economy normalizes. This weighted smoothing approach would allow us to preserve the context of the pandemic while minimizing its distortion on broader trends. Upon researching a bit, I found out that the FB Prophet model actually solves this problem with the same approach.

Facebook Prophet Model

Facebook's Prophet model is designed to handle outliers in time series data gracefully, ensuring they do not unduly influence the overall trend or seasonal components. It uses a robust regression method with a student's t-distribution instead of a normal distribution to model the residuals (errors). This approach makes the model less sensitive to large deviations caused by outliers, as the t-distribution has heavier tails compared to the normal distribution. Facebook's Prophet model automatically identifies potential outliers by analyzing how much a data point deviates from the expected trend. It reduces the weight of outliers during model fitting, effectively downplaying their influence. Regular data points are given higher weights. Outliers are assigned lower weights during optimization.

Forecast by the Model

Date	**Forecast**
2024-11-01	4.503224
2024-12-01	4.230388
2025-01-01	4.154330
2025-02-01	4.262189
2025-03-01	4.047508
2025-04-01	3.930299
2025-05-01	4.003752
2025-06-01	3.946071
2025-07-01	3.912008
2025-08-01	3.711117
2025-09-01	3.662440
2025-10-01	3.482036

As I mentioned earlier, the model uses robust regression with a Student's t-distribution, which is less sensitive to extreme data points (outliers) than a normal distribution. The smoothness and stability of the forecast indicate that any outliers in the historical data did not overly influence the model's identified trends or seasonal patterns. By automatically reducing the weight of such outliers during fitting, the model ensures the predictions reflect the core structure of the data rather than noise or anomalies. This solves our underlying problem with the exact same approach that I had in my mind in the beginning. Because outliers in historical data are down weighted, but their influence is not entirely ignored. This ensures the model remains flexible enough to adapt to large deviations while still focusing on the core trends. Therefore, the model predicts that we are past the recovery by COVID-19 and now the data is going to follow the trend it had before the pandemic and unemployment is going to reduce further down.

Regression Analysis

Regression post 2010 without forecasts:

```
reg Unemployment JobOpenings
```

```
      Source |       SS           df       MS      Number of obs   =       165
-------------+----------------------------------   F(1, 163)       =    103.84
       Model |  273.721029         1  273.721029   Prob > F        =    0.0000
    Residual |  429.651334       163  2.63589776   R-squared       =    0.3892
-------------+----------------------------------   Adj R-squared   =    0.3854
       Total |  703.372364       164  4.28885588   Root MSE        =    1.6235

------------------------------------------------------------------------------
Unemployment |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
 JobOpenings |  -.9983075   .0979658   -10.19   0.000    -1.191753   -.8048618
       _cons |   9.841208    .439749    22.38   0.000     8.972869    10.70955
------------------------------------------------------------------------------
```

Regression with forecasts:

```
reg Unemployment JobOpenings
```

```
      Source |       SS           df       MS      Number of obs   =       177
-------------+----------------------------------   F(1, 175)       =     90.02
       Model |  248.485347         1  248.485347   Prob > F        =    0.0000
    Residual |  483.059745       175   2.7603414   R-squared       =    0.3397
-------------+----------------------------------   Adj R-squared   =    0.3359
       Total |  731.545093       176  4.15650621   Root MSE        =    1.6614

------------------------------------------------------------------------------
Unemployment |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
 JobOpenings |  -.9422862   .0993148    -9.49   0.000    -1.138295   -.7462774
       _cons |   9.454553   .4408455    21.45   0.000     8.584495    10.32461
------------------------------------------------------------------------------
```

Interpretation of Results

As the absolute value of the coefficient decreases, this appears inconsistent with our understanding of the growing importance of job openings. However, this discrepancy may not be a significant issue, as it could be attributed to the reduced influence of outliers in the analysis. During the recovery period, decreases in the unemployment rate were associated with smaller increases in job openings, while increases in the unemployment rate resulted in smaller decreases in job openings. This dynamic may explain the observed trend. Therefore, the behavior of the coefficient can be reasonably interpreted within the framework of the basic model specification.

FED Behaviour

Unemployment is decreasing, reflecting an improving labor market, while a moderate negative correlation indicates job creation is ongoing but with diminishing effects post-pandemic. Volatility from the recovery phase suggests a return to pre-pandemic patterns. The Federal Reserve, under its dual mandate of maximum employment and stable prices, should monitor for wage inflation or labor shortages. If inflation remains controlled, maintaining supportive monetary policy may foster job growth. However, if labor markets overheat, tightening policies like raising interest rates may stabilize the economy. Accounting for pandemic outliers is crucial to avoid misinterpreting short-term volatility and ensure long-term economic stability as the results do change by removing the effects.